

Amazon Customer Analysis Dashboard

Executive Summary

In today’s competitive e-commerce market, understanding customer behavior, sales performance, returns, and seller efficiency is essential for growth. This project focuses on analyzing a large-scale Amazon-style e-commerce dataset containing **1 million customer transaction records**.

The objective was to transform raw transactional data into actionable business insights using an end-to-end analytics workflow involving **Python, SQL, PostgreSQL, and Power BI**.

The final result is an interactive **Customer Analysis Dashboard** that helps businesses monitor revenue, customer orders, return rates, device usage, city performance, and monthly sales trends for smarter decision-making.

Project Objective

The goal of this project was to answer critical business questions such as:

- Which product categories generate the highest revenue?
- Which cities contribute most to sales?
- What is the customer return rate?
- Which devices are customers using to place orders?
- How are monthly sales performing?
- Which sellers perform best based on ratings and revenue?

Tools & Technologies Used

Tool	Purpose
Python	Data cleaning & preprocessing
Pandas	Data manipulation
PostgreSQL	Database storage
SQL	Business analysis queries
Power BI	Dashboard creation
Jupyter Notebook	Development environment

Dataset Overview

The project uses a synthetic Amazon e-commerce dataset with approximately **1,000,000 rows** and multiple business-related columns.

```
[5]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   user_id                1000000 non-null object  
1   product_id             1000000 non-null object  
2   category               1000000 non-null object  
3   subcategory            1000000 non-null object  
4   brand                  1000000 non-null object  
5   price                  1000000 non-null float64  
6   discount               1000000 non-null float64  
7   final_price            1000000 non-null float64  
8   rating                 999967 non-null float64  
9   review_count           1000000 non-null int64  
10  stock                  1000000 non-null int64  
11  seller_id              1000000 non-null object  
12  seller_rating          999975 non-null float64  
13  purchase_date          1000000 non-null datetime64[ns]
14  shipping_time_days     1000000 non-null int64  
15  location               1000000 non-null object  
16  device                 1000000 non-null object  
17  payment_method         1000000 non-null object  
18  is_returned            1000000 non-null bool  
19  delivery_status        1000000 non-null object  
dtypes: bool(1), datetime64[ns](1), float64(5), int64(3), object(10)
memory usage: 145.9+ MB
```

Key Features Included:

- User ID
- Product ID
- Category / Subcategory
- Brand
- Price / Discount / Final Price
- Product Rating
- Seller Rating
- Purchase Date
- Shipping Time
- City
- Device Type
- Payment Method
- Return Status
- Delivery Status

This dataset simulates real-world e-commerce transactions and provides valuable analytical opportunities.

Data Cleaning & Preparation (Python)

Before analysis, raw data was cleaned and standardized using Python.

```
[7]: df.isnull().sum()

[7]: user_id            0
     product_id       0
     category         0
     subcategory      0
     brand            0
     price            0
     discount         0
     final_price      0
     rating           33
     review_count     0
     stock           0
     seller_id        0
     seller_rating    25
     purchase_date    0
     shipping_time_days 0
     location         0
     device           0
     payment_method   0
     is_returned      0
     delivery_status  0
     dtype: int64

[8]: df['rating'] = df.groupby(['category', 'subcategory', 'brand'])['rating'].transform(lambda x: x.fillna(x.median()))

[9]: df['seller_rating'] = df.groupby(['category', 'subcategory', 'brand'])['seller_rating'].transform(lambda x: x.fillna(x.median()))

10]: df.isnull().sum()

10]: user_id            0
     product_id       0
     category         0
     subcategory      0
     brand            0
     price            0
     discount         0
     final_price      0
     rating           0
     review_count     0
     stock           0
     seller_id        0
     seller_rating     0
     purchase_date    0
     shipping_time_days 0
     location         0
     device           0
     payment_method   0
```

Key Steps Performed:

- Loaded CSV data using Pandas
- Handled missing values
- Filled missing seller ratings using median values
- Renamed inconsistent columns
- Converted dates into proper datetime format
- Checked duplicates and null records
- Exported clean dataset for SQL and Power BI usage

Example Skills Demonstrated:

- Pandas DataFrames
- Data wrangling
- Data quality checks
- ETL workflow basics

SQL Analysis (PostgreSQL)

Structured Query Language (SQL) was used to derive business insights from the cleaned dataset.

Example Analysis Performed:

- Top cities by total revenue
- Monthly revenue trends
- Return rate by category
- Top-performing sellers
- Product stock risk analysis
- Ranking categories using window functions
- Revenue contribution analysis

```
3 --1. Top 5 Cities by Gross Revenue
4 SELECT city, ROUND(SUM(final_price)::numeric,2) AS gross_revenue, RANK() OVER (ORDER BY SUM(final_price) DESC) AS ranking
5 GROUP BY city;
6
7 --2. Monthly Trend
8 SELECT EXTRACT (month FROM purchase_date) AS month_purchase_date, ROUND(SUM(final_price)::numeric,2) FROM customer
9 GROUP BY month_purchase_date;
10
11 --3. Return Rate by Category
12 SELECT category, ROUND(AVG(CASE WHEN is_returned = true THEN 1 ELSE 0 END)*100,2) AS return_rate FROM customer
13 GROUP BY category
14 ORDER BY return_rate DESC;
15
16 --4. Product Selling above Average Product Price
17 SELECT product_id, category, final_price FROM (SELECT product_id,category,final_price, AVG(final_price) OVER(PARTITION B
18 WHERE final_price > avg_price;
19
20 --5. Device wise Customer Spending
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	city text	gross_revenue numeric	ranking bigint
1	Delhi	2000521360.30	1
2	Bangalore	1990173998.60	2
3	Mumbai	1988987898.72	3
4	Chennai	1981129736.81	4
5	Hyderab...	1978063990.47	5

SQL Concepts Used:

- GROUP BY
- ORDER BY
- CASE WHEN
- CTEs
- Window Functions
- RANK()
- ROW_NUMBER()
- Aggregate Functions

```
40 ORDER BY returns DESC;
41
42 --9. Products at Risk: High Demand but Low Stock
43 SELECT product_id,category,stock,review_count,rating,
44         CASE
45             WHEN stock < 50 AND review_count > 30 AND rating >= 4
46             THEN 'Restock Urgently'
47             WHEN stock < 100
48             THEN 'Monitor'
49             ELSE 'Healthy'
50         END AS stock_status
51 FROM customer
52 ORDER BY stock ASC, review_count DESC;
53
54 --10. Best Performing Payment Method
55 SELECT payment_method,ROUND(SUM(final_price)::numeric,2) AS revenue,COUNT(*) AS total_orders FROM customer
56 GROUP BY payment_method
57 ORDER BY revenue DESC;
```

Data OutputMessagesNotifications

SQL

Showing rows: 1 to 1000Page No: 1 of 1000

	product_id text	category text	stock bigint	review_count bigint	rating double precision	stock_status text
1	P98237	Home	0	508	3.5	Monitor
2	P33567	Home	0	435	4.2	Restock Urgen...
3	P32564	Electroni...	0	427	4.2	Restock Urgen...
4	P15402	Home	0	408	3	Monitor
5	P93483	Clothing	0	366	4	Restock Urgen...

Total rows: 1000000Query complete 00:00:02.213CRLF Ln 53, Col 1

Power BI Dashboard Overview

An interactive dashboard was developed to visualize business KPIs and trends.

KPI Cards

- Total Orders: **1M**
- Net Revenue: **₹8.82 Billion**
- Return Rate: **11.60%**
- Average Order Value: **₹9.94K**
- Average Seller Rating: **3.75**

Dashboard Visualizations

1. Net Revenue by Category

Tracks the highest earning categories.

2. Return Rate by Category

Identifies categories with operational issues or product dissatisfaction.

3. Orders by Device Type

Shows customer purchase behavior across Mobile App, Web, and Tablet.

4. Revenue by City

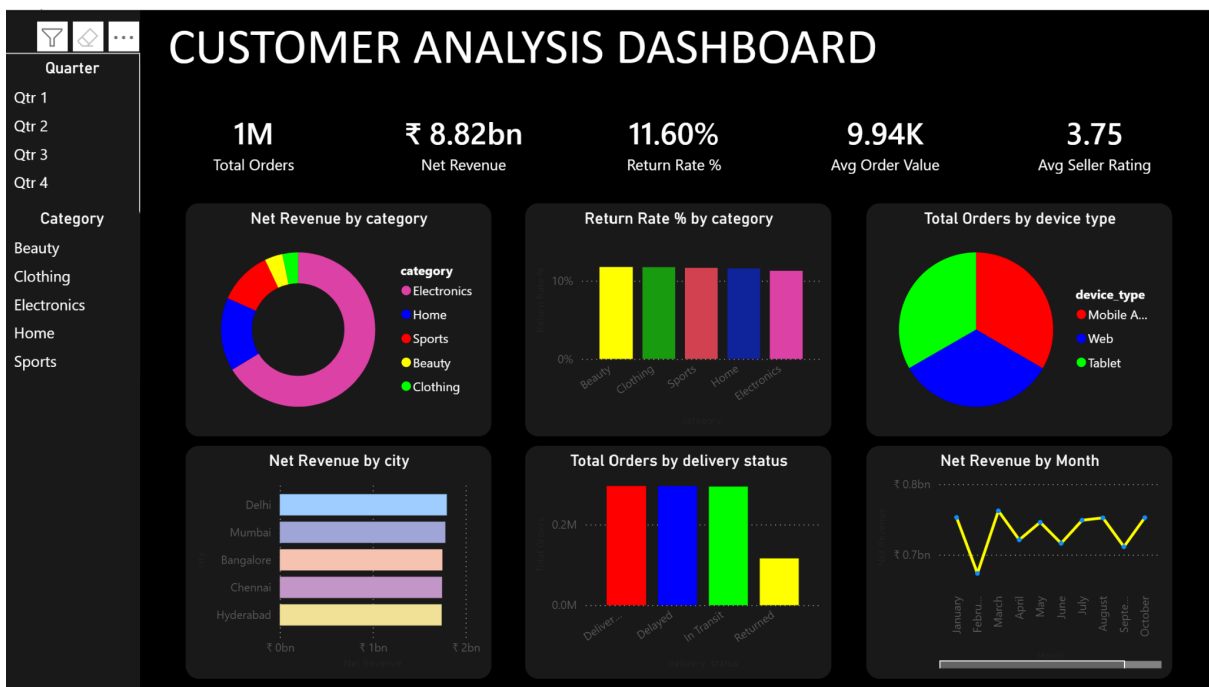
Compares city-wise sales performance.

5. Orders by Delivery Status

Monitors logistics efficiency.

6. Monthly Revenue Trend

Tracks seasonality and growth patterns.



Key Business Insights

Revenue Performance

Electronics emerged as the highest revenue-generating category, indicating strong customer demand.

Return Rate

Return rate of **11.6%** suggests opportunities to improve product quality, expectations, or delivery handling.

Device Usage

Orders are distributed across multiple devices, showing the importance of omnichannel experience.

Geographic Trends

Major metro cities contribute significantly to revenue.

Seller Quality

Average seller rating of **3.75** indicates room for improving marketplace quality.

Business Recommendations

Improve Return Management

Investigate categories with higher return percentages.

Optimize Seller Standards

Introduce stricter seller quality benchmarks.

Focus on Top Cities

Run targeted campaigns in best-performing cities.

Mobile Optimization

Enhance mobile shopping experience if mobile contributes highest traffic.

Seasonal Marketing

Use monthly sales trends for promotions and inventory planning.

Conclusion

This project successfully converts large-scale e-commerce data into actionable insights using modern analytics tools. It showcases both technical ability and business thinking, making it a strong portfolio project for data analytics opportunities.